SOFTWARE ENGINEERING LAB 1

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SECTION G

PROBLEM STATEMENT

Hospital Bed & ICU Allocation Dashboard

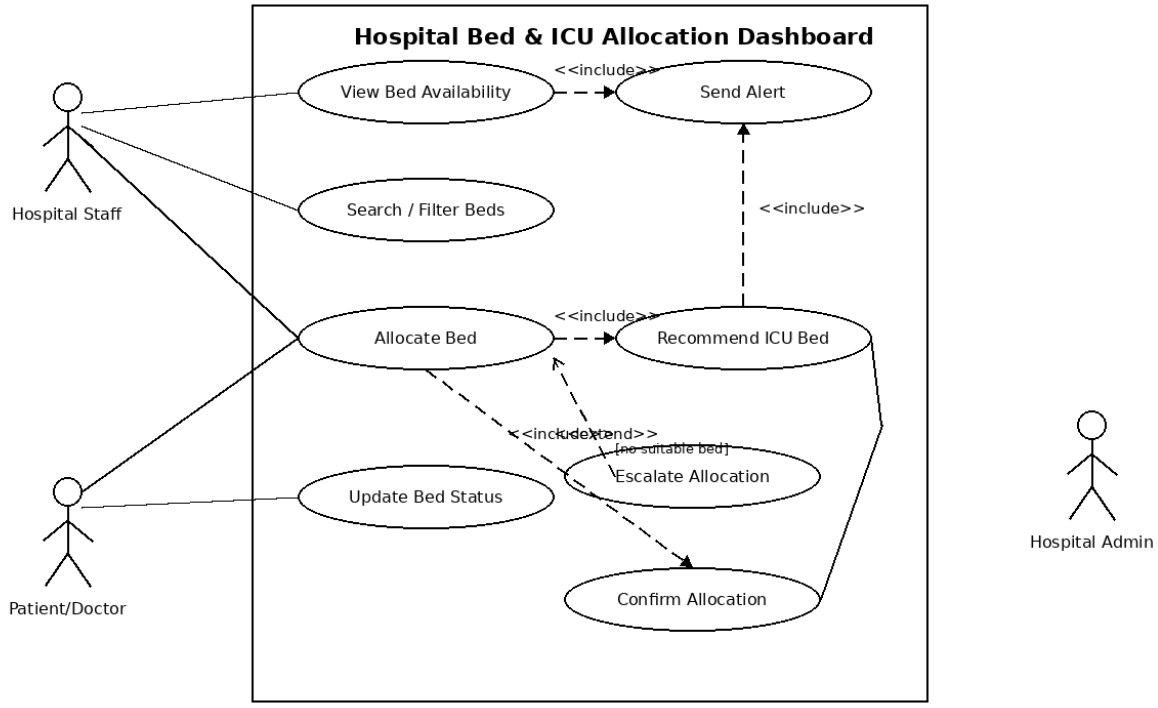
1. Requirements Table :

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE	COMMENTS
FR-001	Functional	The system shall display real-time bed availability across wards, including general, emergency, and ICU beds, with occupied, available, reserved, and cleaning status.	High	Pass: dashboard shows an accurate current count for each bed status. Fail: an occupied or reserved bed is shown as available.	Real-time visibility is the core purpose of the allocation dashboard.	Core dashboard requirement
FR-002	Functional	The system shall allow authorized hospital staff to search and filter beds by ward, bed type, ICU availability, and current status.	High	Pass: filters return only beds matching the selected criteria. Fail: unavailable beds appear in the available result set.	Fast bed discovery supports efficient patient placement.	Supports efficient bed discovery
FR-003	Functional	The system shall allow hospital staff to allocate an available bed to a patient by recording patient details, required ward/type, and admission priority.	High	Pass: a selected available bed is assigned to the patient and its status changes to occupied/reserved. Fail: two patients can be assigned the same bed.	Prevents allocation conflicts and supports admission workflow.	Prevents duplicate allocation
FR-004	Functional	The system shall recommend suitable ICU beds for critical patients based on ICU type, equipment requirements, and bed availability.	High	Pass: recommendations include only compatible available ICU beds. Fail: a non-ICU or incompatible bed is recommended for an ICU requirement.	Supports safe and appropriate ICU allocation during emergencies.	Supports safe ICU placement
FR-005	Functional	The system shall allow hospital administrators to update bed status, release beds, and view alerts for low ICU capacity or allocation conflicts.	High	Pass: status changes are reflected on the dashboard immediately and conflicts generate an alert. Fail: released beds remain unavailable.	Accurate status management keeps the dashboard reliable.	Maintains accurate capacity status
NFR-001	Nonfunctional, Performance	The dashboard shall load current bed and ICU availability within 2 seconds under a load of 500 concurrent users.	High	Pass: load testing confirms dashboard response under 2 seconds at 500 concurrent users. Fail: response exceeds 2 seconds or availability data is dropped.	Rapid updates are critical for admission and emergency decisions.	Measurable performance target
NFR-002	Nonfunctional, Security	All patient, admission, and bed-allocation data shall be encrypted in transit and at rest, with access restricted to authorized hospital users.	High	Pass: security review confirms TLS in transit, encrypted storage, and no sensitive patient data in plaintext logs. Fail: unauthorized users can access protected data.	Protects sensitive patient and hospital operational data.	Protects sensitive patient data

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2. Use Case Diagram :



3. Use-Case Flow Document :

Use Case: Allocate Bed / ICU Bed

Actors: Hospital Staff, Bed Allocation System, Hospital Admin

Preconditions: The hospital staff has registered and is logged in. The bed and ICU allocation system is operational and the availability data is up to date.

Postconditions: The patient is assigned a suitable available bed or ICU bed. The dashboard is updated accordingly, and the hospital staff receives a confirmation notification.

Main Success Scenario:

1. Hospital staff logs in and selects the option to allocate a bed.
2. Hospital staff enters the patient details, required ward/bed type, and admission priority, then submits the request.
3. System checks current bed and ICU availability based on the requested requirements.
4. If a suitable bed is available, the system reserves the bed and skips to step 9.
5. If an ICU bed is required, the system recommends compatible available ICU beds based on ICU type and equipment requirements.
6. System displays the suitable bed options to the hospital staff.
7. Hospital staff selects the appropriate bed and submits the allocation for confirmation.
8. Hospital admin verifies the allocation when required and confirms the assignment.
9. System marks the selected bed as occupied/reserved and updates the dashboard.

10. System sends a confirmation notification to the hospital staff.

11. Use case ends successfully.

Alternate Flow:

4a. No Suitable Bed Available

4a1. System records the patient as waiting and removes unavailable beds from the active allocation list.

4a2. System sends an alert to the hospital admin when ICU or ward capacity is critically low.

4a3. If no suitable bed becomes available, the request remains in the waiting list and hospital staff is notified of the delay. This condition triggers the «extend» use case Escalate Allocation.